Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **AI-enabled operational workflows**, **customer-facing web platforms**, and API-driven systems, with strong experience in troubleshooting, implementations, stakeholder communication, and fast iteration in production.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and operated a multi-tenant platform for 500+ users, reducing latency from **1.2s to 350ms** while improving reliability, supportability, and stakeholder visibility across user-facing workflows.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving operational reliability and accelerating issue resolution across production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed full-stack React and Flask workflows handling 10,000+ daily requests, translating instructor feedback into maintainable interfaces, backend improvements, and operational enhancements validated directly in production.
- Built AI-assisted grading and rubric workflows with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting clearer handoffs and better operational outcomes for end users.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python experimentation pipelines across 100,000+ records, enabling reproducible evaluation and technical artifacts that improved debugging quality, implementation decisions, and communication across collaborators.
- Produced technical reports and validation scenarios translating findings into actionable guidance, strengthening cross-team alignment, reducing handoff ambiguity, and supporting better real-world engineering decisions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting distributed integrations, structured delivery, and reliable customer-facing operations.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services, responsive interfaces, and maintainable integrations.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during implementation, validation, and troubleshooting.

Skills

Solutions & Delivery: client-facing troubleshooting, implementations, stakeholder communication, requirements translation, operational debugging
Web & APIs: JavaScript/TypeScript, React, Next.js, Node.js, Flask, FastAPI, REST APIs, WebSockets
AI & Automation: LLM applications, OCR workflows, prompt-based systems, AI-enabled operations, process automation
Platforms: PostgreSQL, MySQL, MongoDB, AWS, Docker, Git/GitHub, Jenkins, CI/CD, cross-functional collaboration

Projects

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an AI-powered operations-style SaaS platform for image editing and search, integrating scalable storage, responsive interfaces, and premium billing while improving conversion by **18%**.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document platform with semantic search, summarization, and collaborative Q&A, demonstrating AI-agent-style workflows over 50K+ documents while reducing analysis time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, creating prototype-rich workflows for schema exploration, query generation, and faster onboarding into complex operational systems.